Generative AI Application Development using MLflow

Pre-requisite

Familiarity with NLP terms

Transformer

Prompt

LLM

Few shot learning

Eval

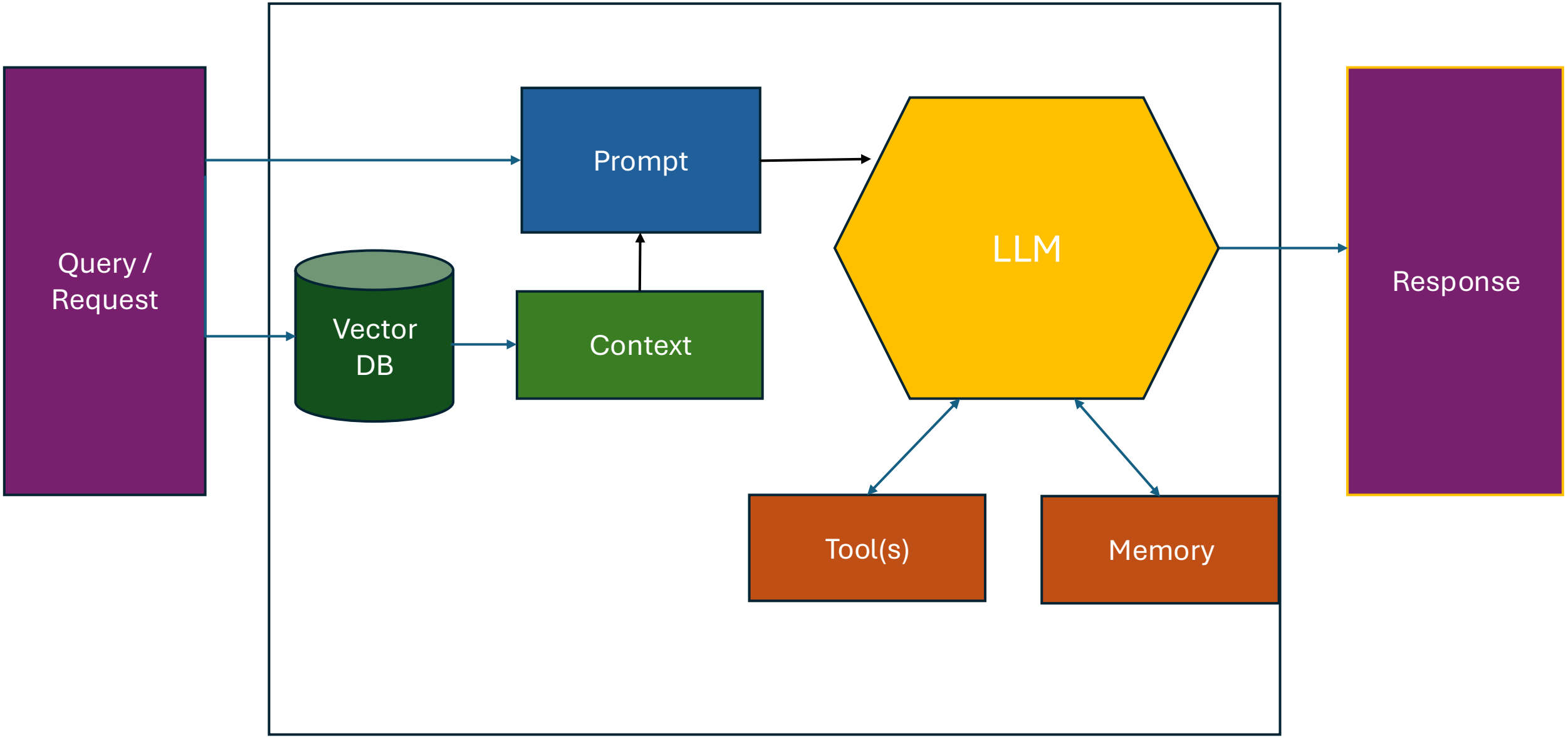
LLM-as-a Judge

Sessions require no prior MLflow familiarity.

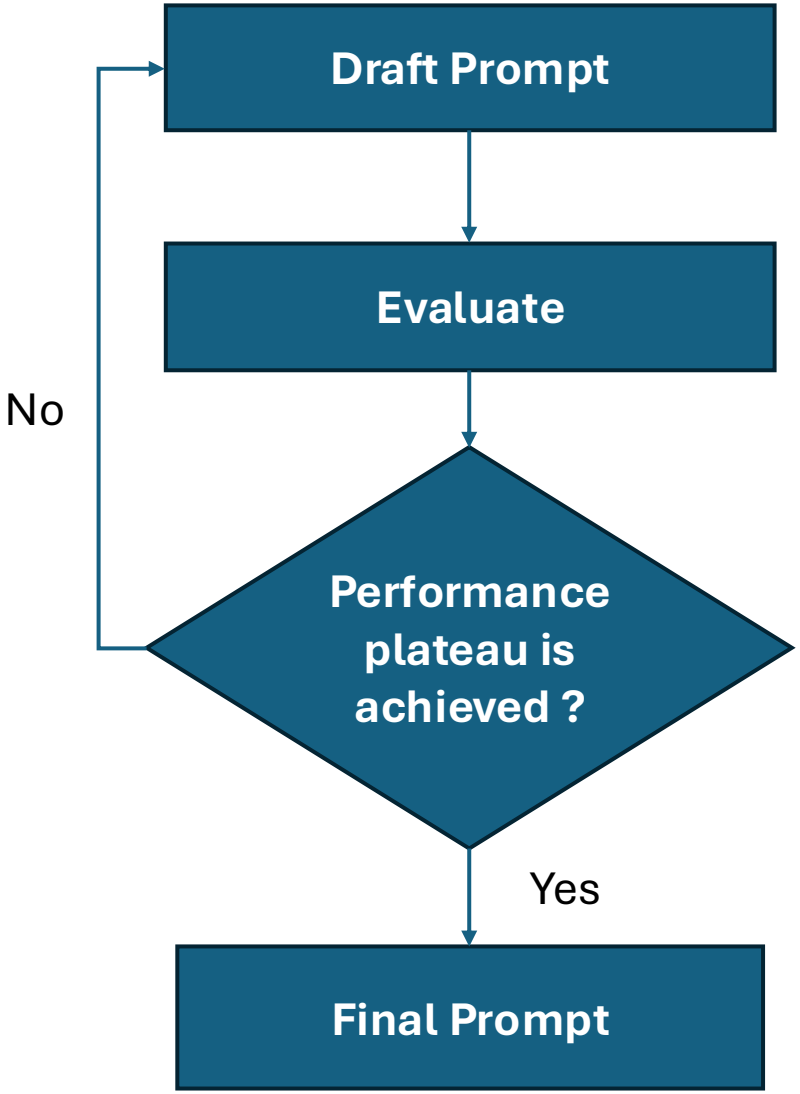
Session Flow

- Revisit Generative AI Basics
- GenAI App Development – Traditional Approach
- GenAI App Development – Leveraging MLflow
- MLflow Features
- Demo
  - Notebook
  - Visual Studio Project
  - Deployment & Hosting

Generative AI Applications (Vanilla →RAG→ Agentic)



GenAI Application Development: Traditional Approach



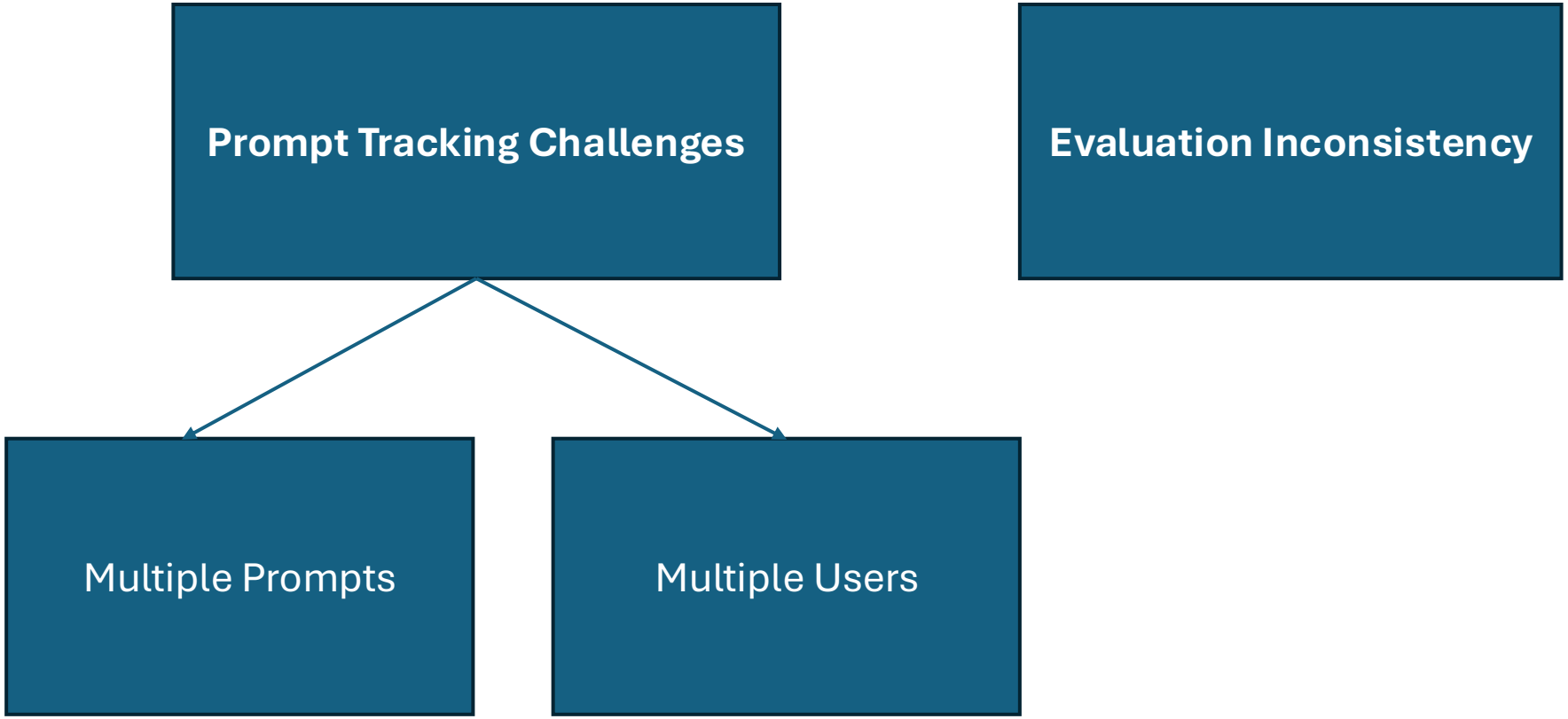
Eval

- Metric
- Dataset

Ground-Truth

Golden-Records

Traditional Approach : Issues



Solution - Experiment Tracking tools

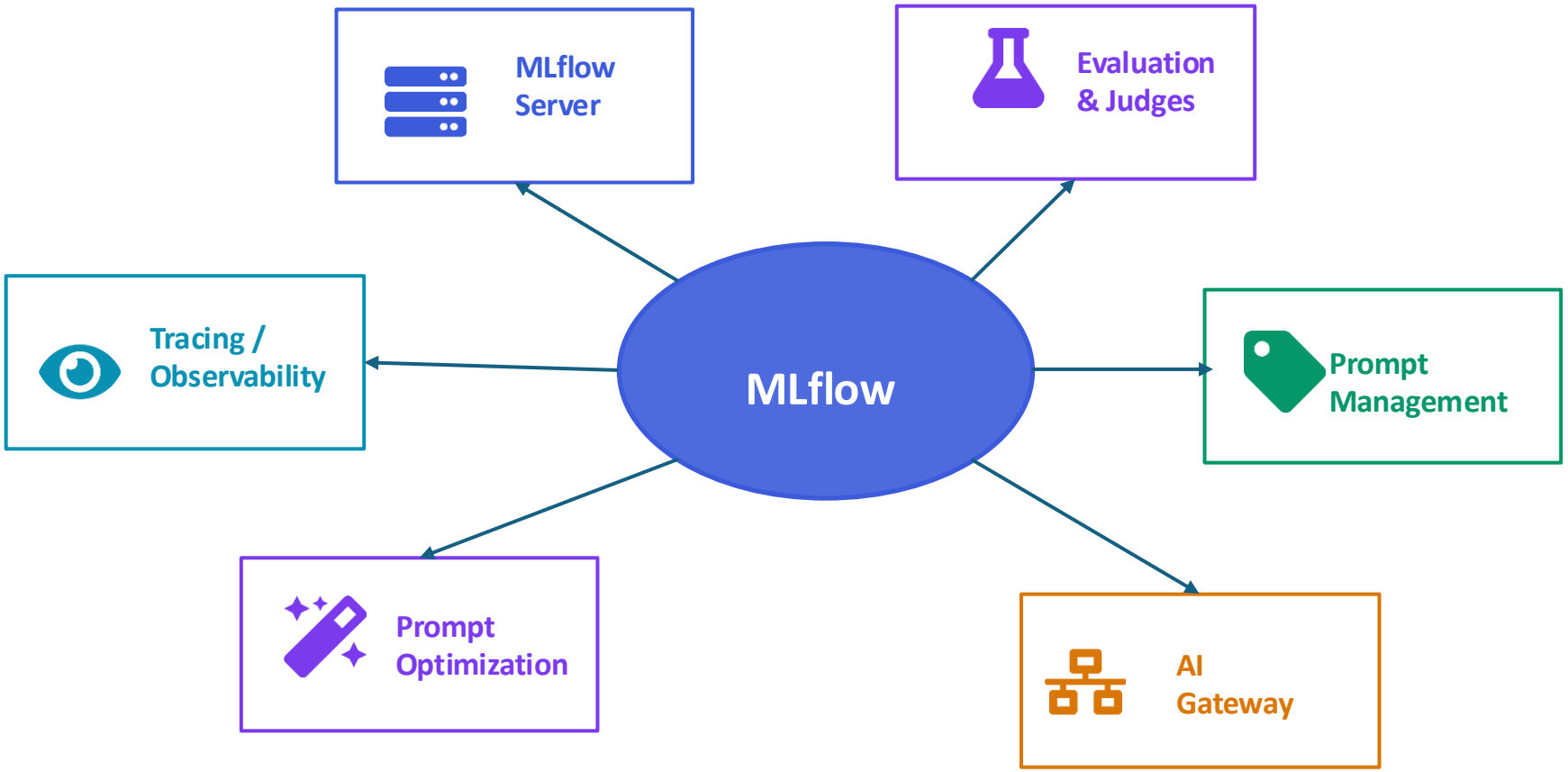
Weights & Biases

mlflow™

Langfuse

- Opensource
- Native GenAI Support
- Feature Rich

MLflow — AI Engineering Platform for Agents & LLMs



One platform — track prompts, evaluate quality, deploy agents, monitor production

Tracing Observability

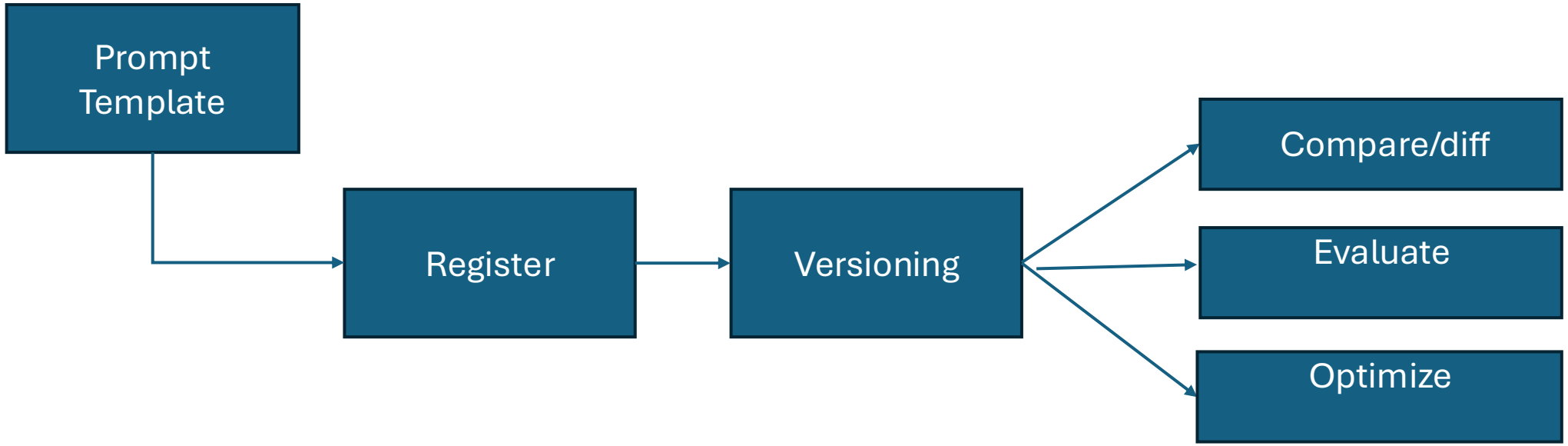
Captures

- Step-by-step execution sequence
- inputs
- outputs
- Metadata

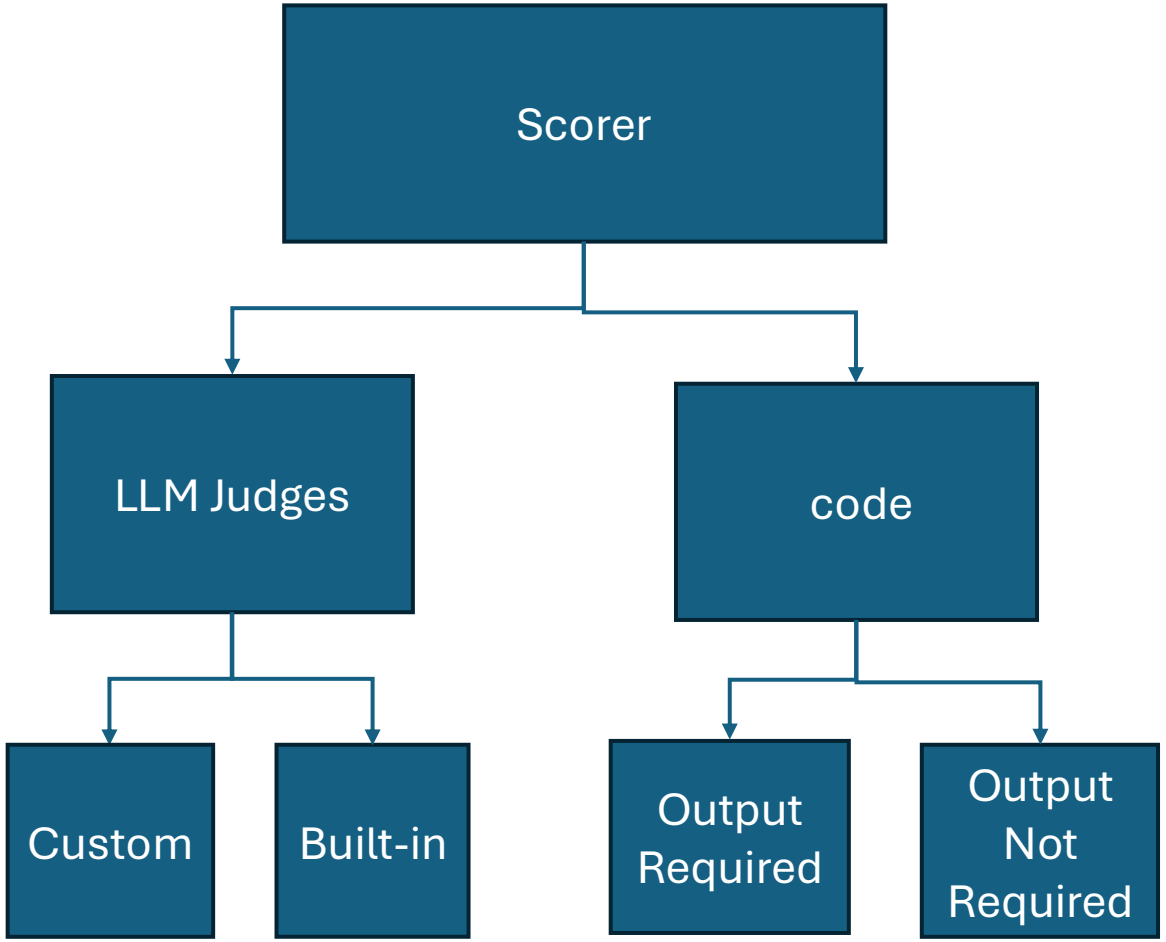
Benefits

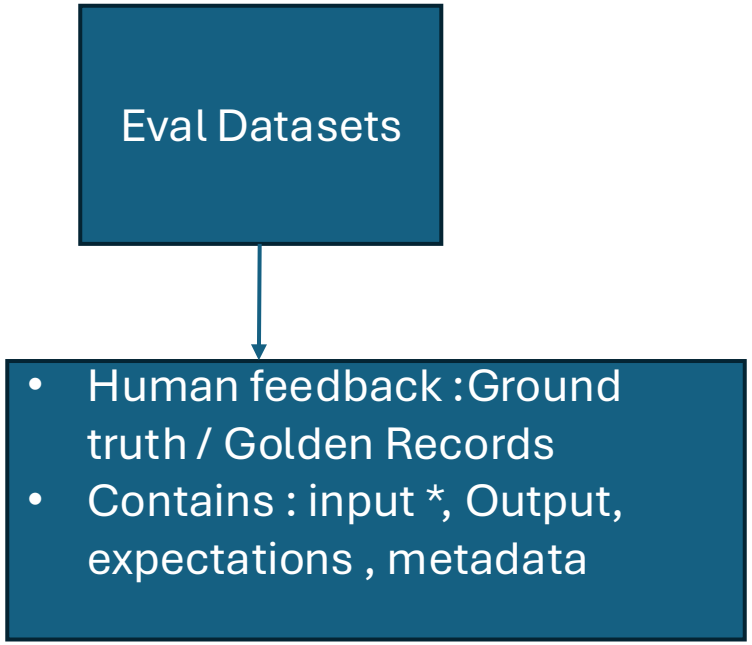
- Debugging
- Production monitoring
- Evaluation
- Dataset Collection

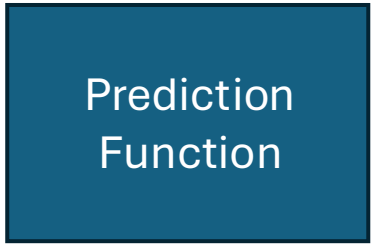
Prompt Management



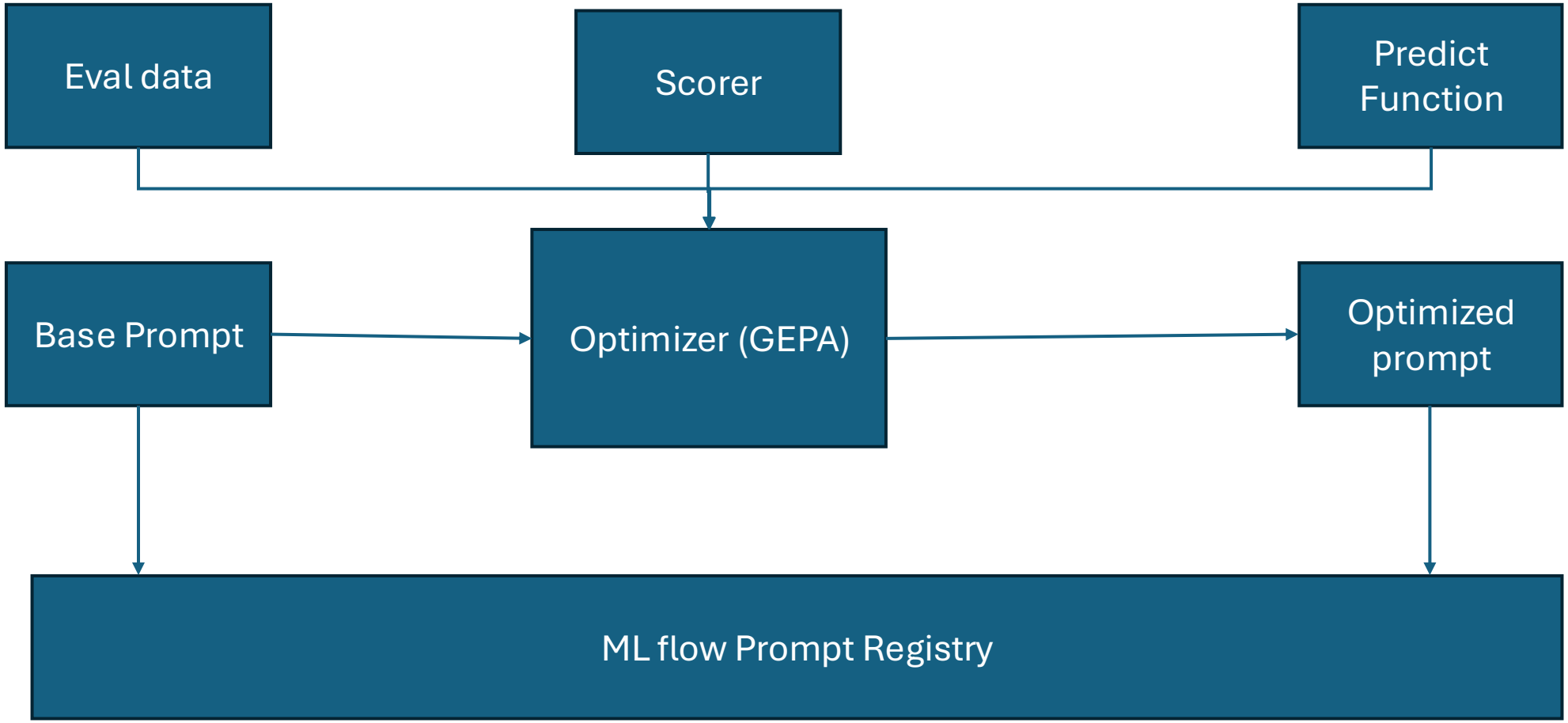
Eval


Scorer
LLM Judges
code
Custom
Built-in
Output Required
Output Not Required


Eval Datasets
• Human feedback :Ground truth / Golden Records
• Contains : input *, Output, expectations , metadata


Prediction Function

Prompt Optimization



Optimization Process

Evaluates current prompt on training examples

Analyse failure patterns

Generates improved prompt variations

Tests these variations to find best performer

Iterates until reach the maximum metric calls or convergence

Demo

https://github.com/kaustuvkunal/genai-support-ticket-classifier-mlflow-demo

Optimization Disclaimers

No guaranteed improvement -Prompt optimization does not always yield better results

Eval data errors propagate - Flaws or biases in evaluation data directly affect the optimized prompt

High compute cost & time -Requires significantly more LLM calls and longer processing time

Inconsistent evaluation results- Scores can vary across runs due to LLM stochasticity

Verbose optimized prompts -Often produces longer and more complex prompts

Q & A